

# TRI-TIERED LOCAL LLM ACTIVE LEARNING FRAMEWORK

*Empirical Evaluation & Cost-Efficient Routing Report*

## Executive Research Summary

This research report compiles results from the empirical evaluation of the Tri-Tiered Active Learning Router. The tiered framework dynamically routes incoming text classification queries to appropriate computation layers (Autonomous DistilBERT, Assisted Qwen2.5:3B LLM, or Human Expert Annotator) based on uncertainty and entropy thresholds. By leveraging dynamic validation threshold sweeps and experience replay active learning, the framework achieves significant reductions in human expert workloads while maximizing model accuracy and utility.

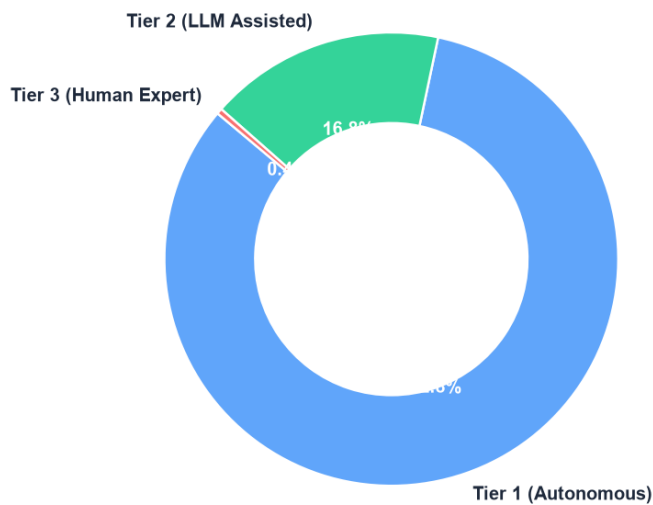
| Dataset    | T1 Acc (Pre) | T1 Acc (Post) | Final Acc | Human Effort % | Net Util (Human) | Net Util (Cost) | Avg Cost |
|------------|--------------|---------------|-----------|----------------|------------------|-----------------|----------|
| AG_NEWS    | 87.40%       | 87.60%        | 89.20%    | 0.40%          | 0.0158           | 0.0152          | 46.43    |
| DBPEDIA_14 | 97.60%       | 97.90%        | 98.30%    | 0.20%          | 0.0039           | 0.0037          | 20.13    |
| EMOTION    | 58.80%       | 60.20%        | 62.80%    | 1.00%          | 0.0255           | 0.0222          | 226.81   |
| IMDB       | 79.40%       | 79.40%        | 87.40%    | 3.60%          | 0.0782           | 0.0767          | 198.51   |

# Dataset Analysis: AG\_NEWS

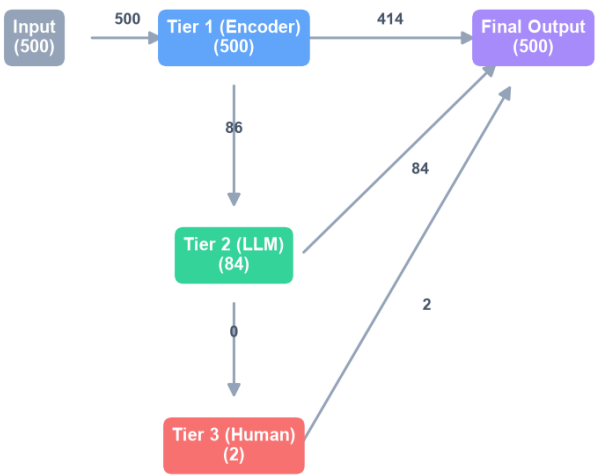
## Performance & Budget Summary Metrics

|                               |        |                              |             |
|-------------------------------|--------|------------------------------|-------------|
| Total Evaluated Samples:      | 500    | Human Effort Ratio:          | 0.40%       |
| Pre-AL T1 Accuracy:           | 87.40% | Post-AL T1 Accuracy:         | 87.60%      |
| Routed Final System Accuracy: | 89.20% | Final System F1 Macro:       | 0.8931      |
| Brier Score (Final):          | 0.2045 | Final System ECE:            | 0.0564      |
| Net Utility (Human-Effort):   | 0.0158 | Net Utility (Cost-Aware):    | 0.0152      |
| Average Compute Cost:         | 46.43  | Active Learning Escalations: | 4 sample(s) |

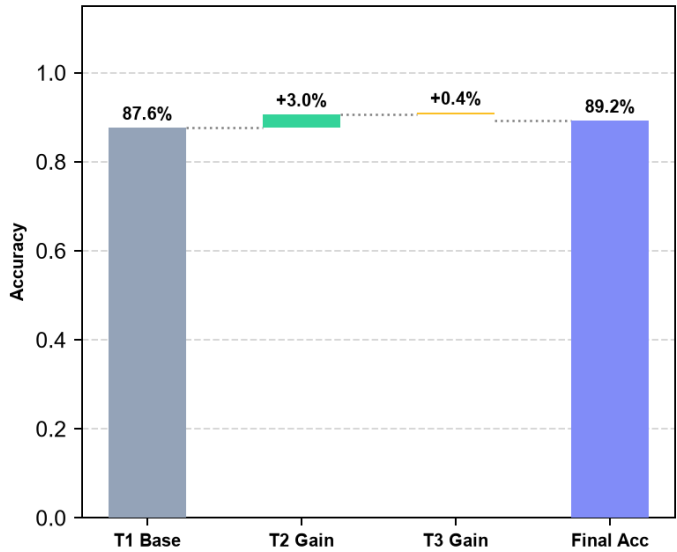
Request Routing Distribution



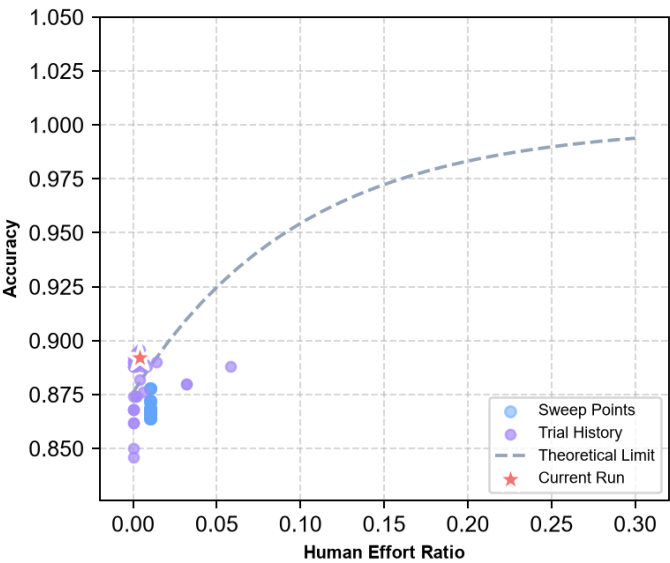
Request Flow Pipeline (Sankey)



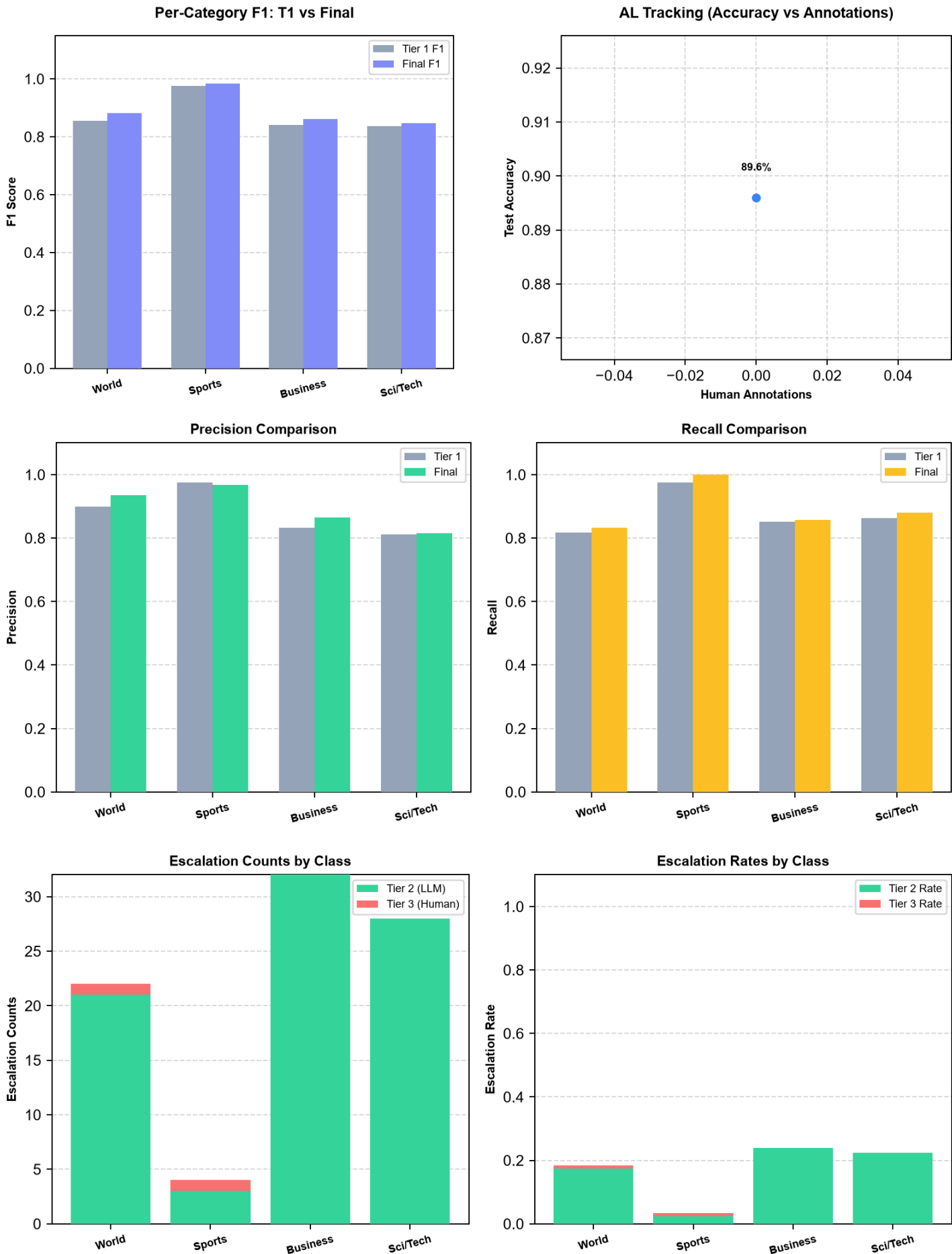
Accuracy Waterfall Study



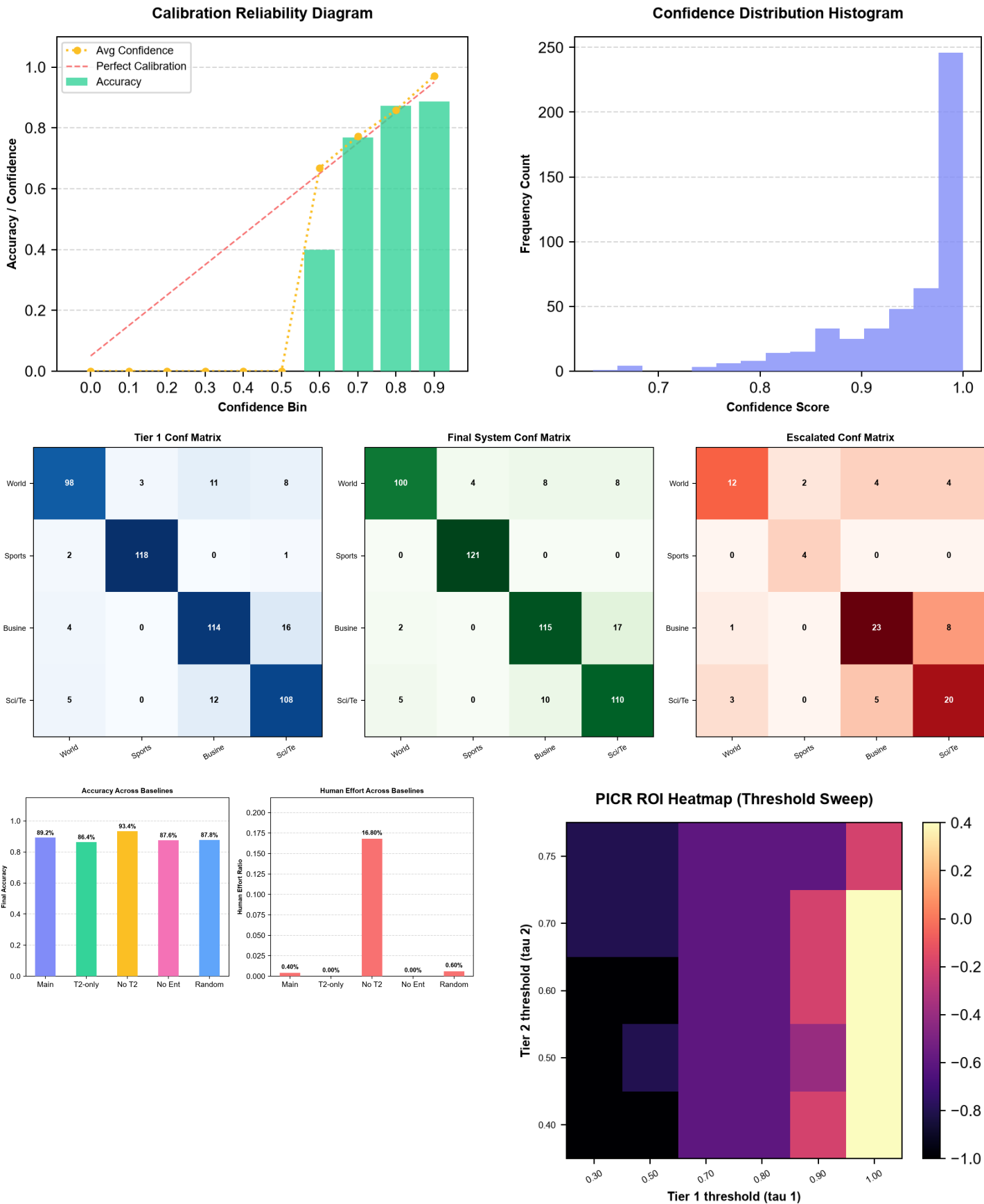
Efficiency Frontier Analysis



# Category & Escalation Profile: AG\_NEWS



# Calibration & Validation Studies: AG\_NEWS

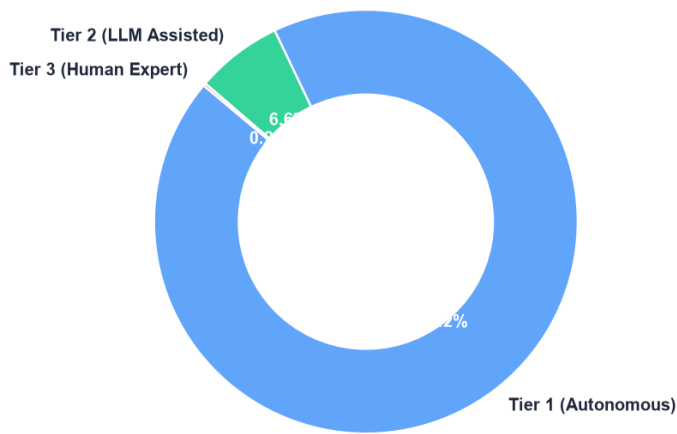


# Dataset Analysis: DBPEDIA\_14

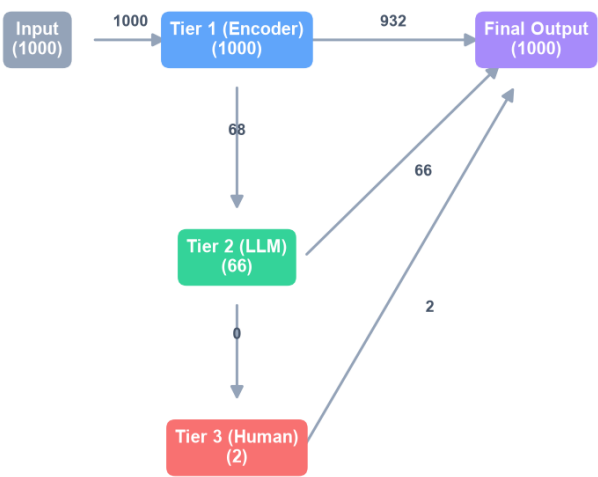
## Performance & Budget Summary Metrics

|                               |        |                              |             |
|-------------------------------|--------|------------------------------|-------------|
| Total Evaluated Samples:      | 1000   | Human Effort Ratio:          | 0.20%       |
| Pre-AL T1 Accuracy:           | 97.60% | Post-AL T1 Accuracy:         | 97.90%      |
| Routed Final System Accuracy: | 98.30% | Final System F1 Macro:       | 0.9835      |
| Brier Score (Final):          | 0.0353 | Final System ECE:            | 0.0143      |
| Net Utility (Human-Effort):   | 0.0039 | Net Utility (Cost-Aware):    | 0.0037      |
| Average Compute Cost:         | 20.13  | Active Learning Escalations: | 2 sample(s) |

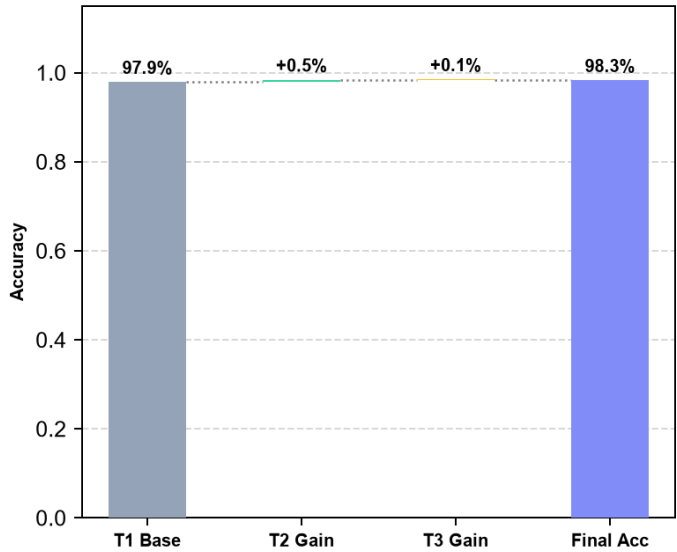
Request Routing Distribution



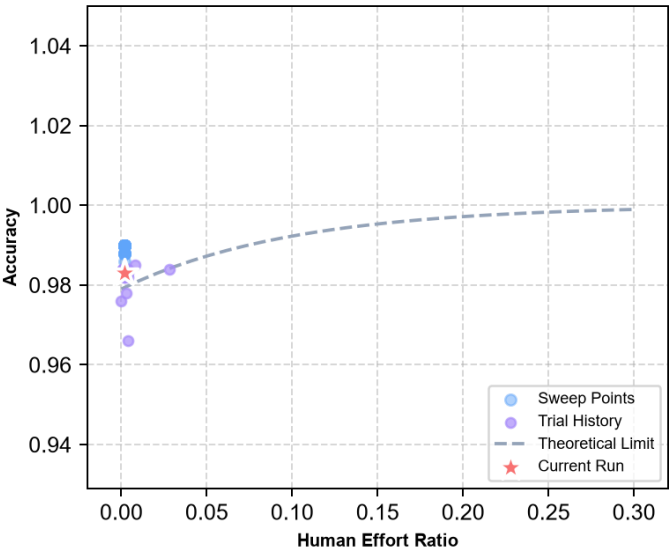
Request Flow Pipeline (Sankey)



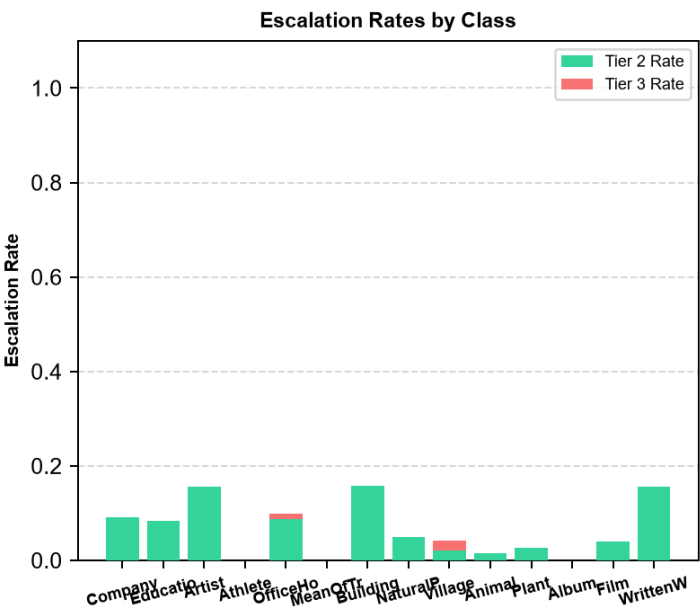
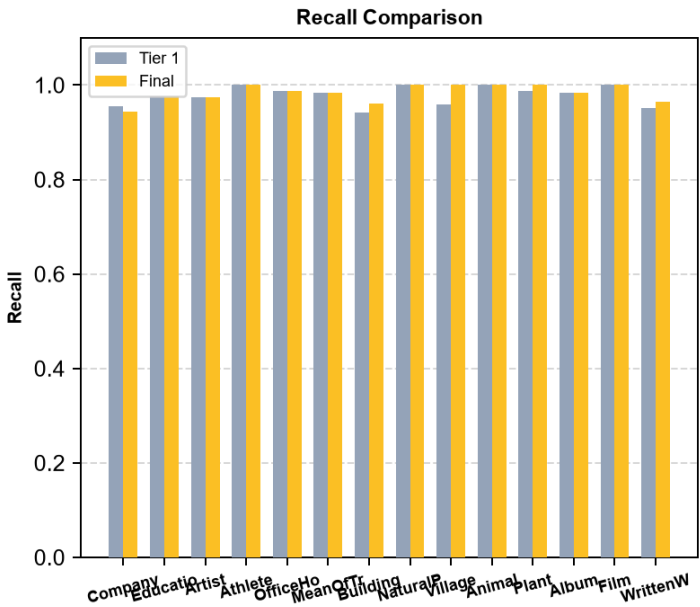
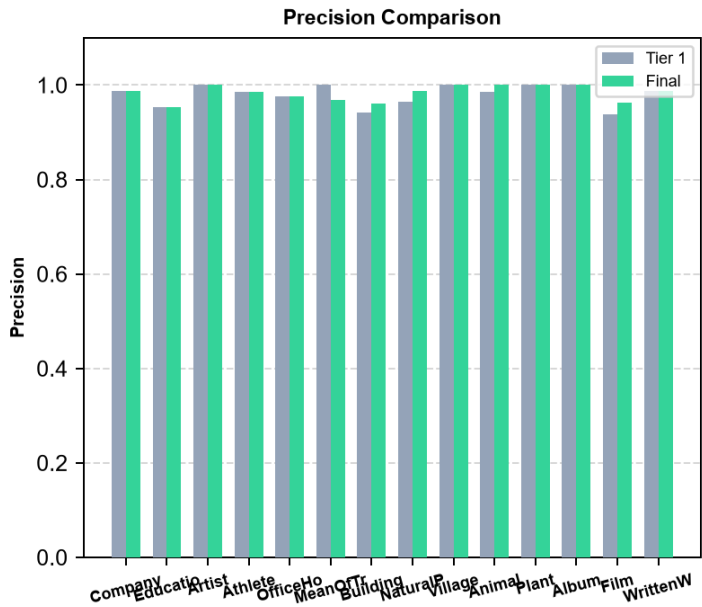
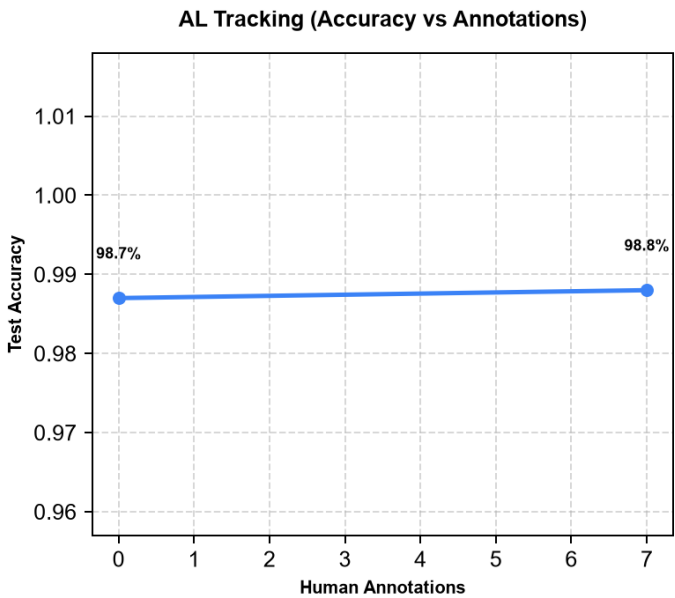
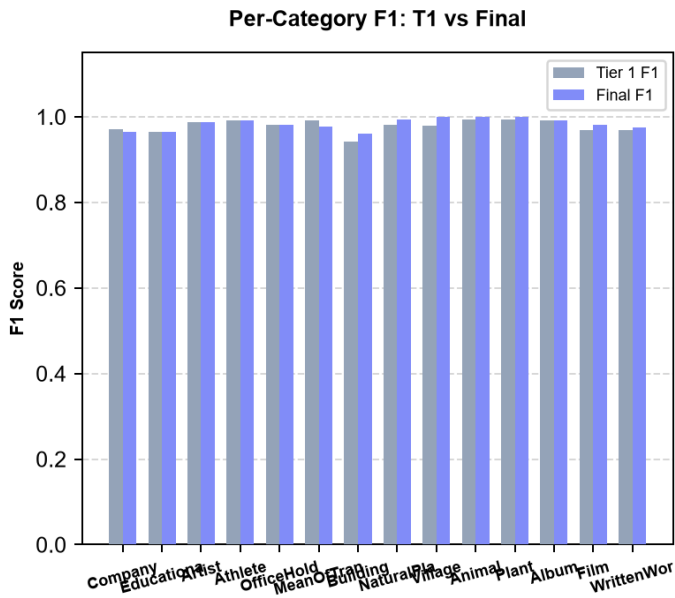
Accuracy Waterfall Study



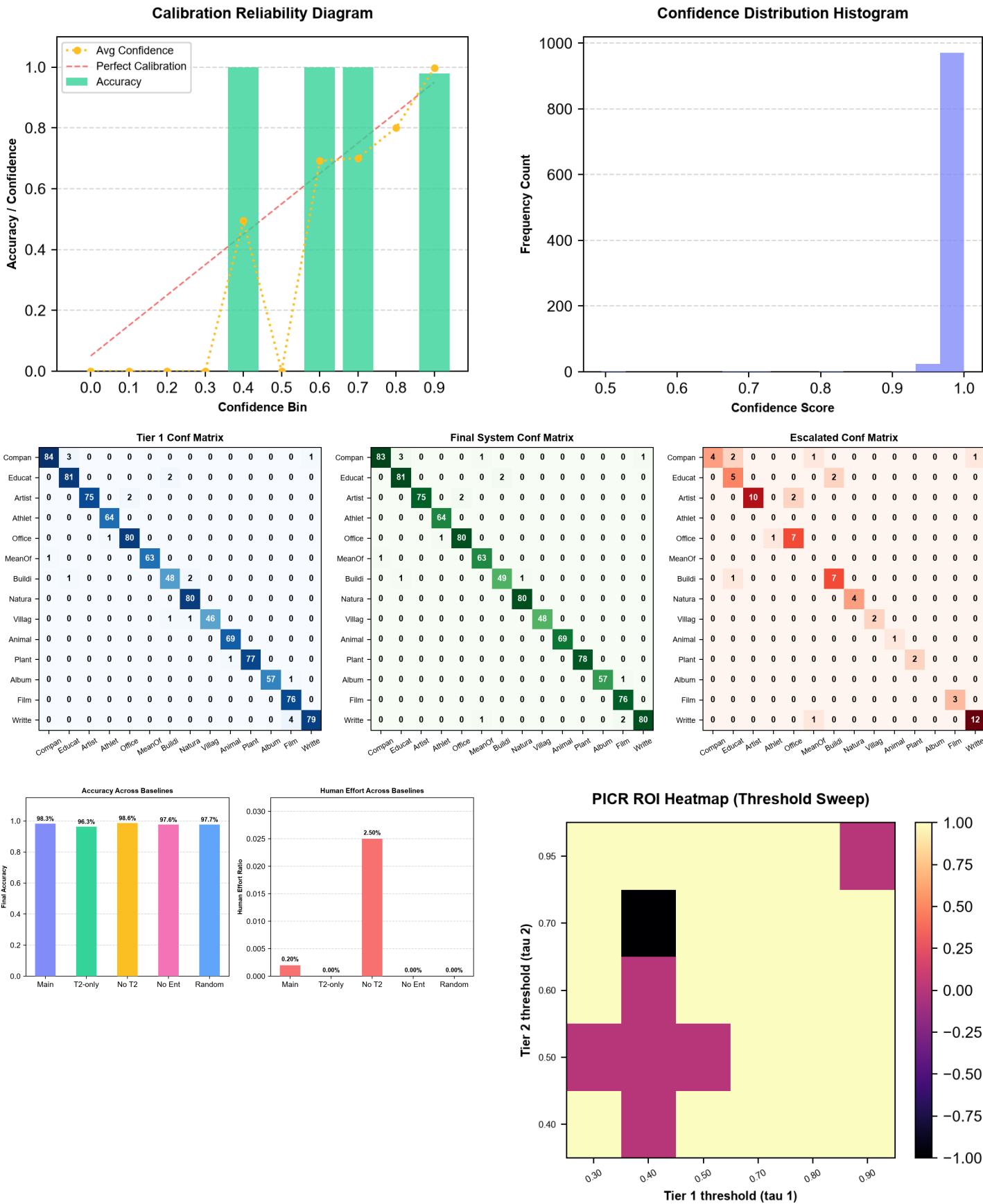
Efficiency Frontier Analysis



# Category & Escalation Profile: DBPEDIA\_14



# Calibration & Validation Studies: DBPEDIA\_14

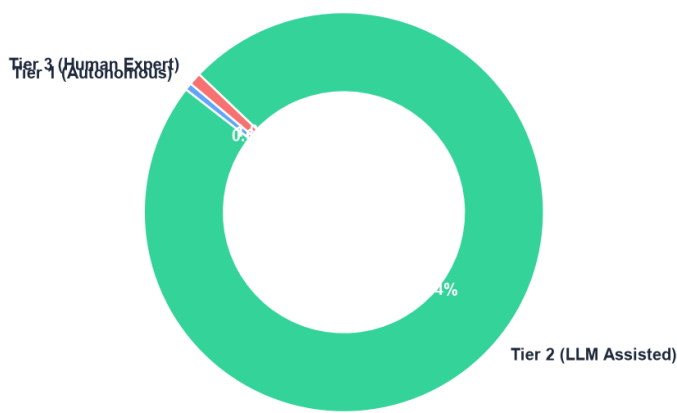


# Dataset Analysis: EMOTION

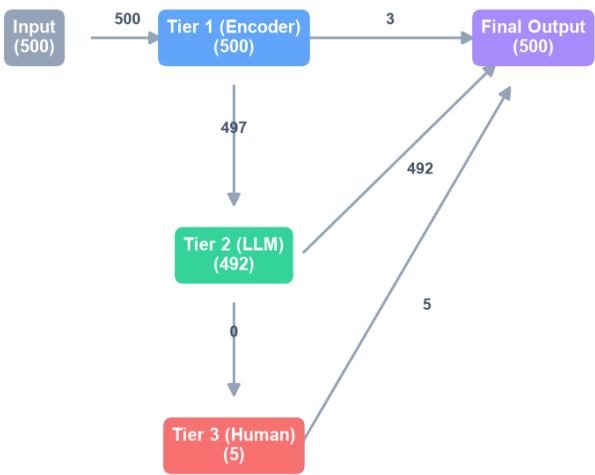
## Performance & Budget Summary Metrics

|                               |        |                              |              |
|-------------------------------|--------|------------------------------|--------------|
| Total Evaluated Samples:      | 500    | Human Effort Ratio:          | 1.00%        |
| Pre-AL T1 Accuracy:           | 58.80% | Post-AL T1 Accuracy:         | 60.20%       |
| Routed Final System Accuracy: | 62.80% | Final System F1 Macro:       | 0.3994       |
| Brier Score (Final):          | 0.7066 | Final System ECE:            | 0.3067       |
| Net Utility (Human-Effort):   | 0.0255 | Net Utility (Cost-Aware):    | 0.0222       |
| Average Compute Cost:         | 226.81 | Active Learning Escalations: | 22 sample(s) |

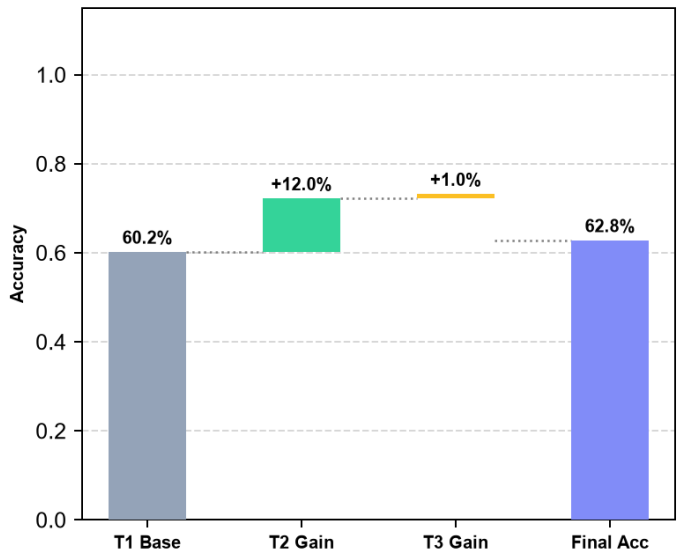
Request Routing Distribution



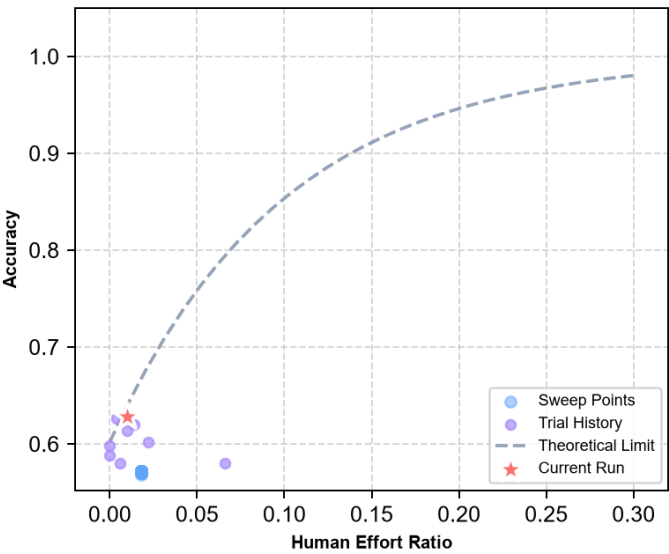
Request Flow Pipeline (Sankey)



Accuracy Waterfall Study

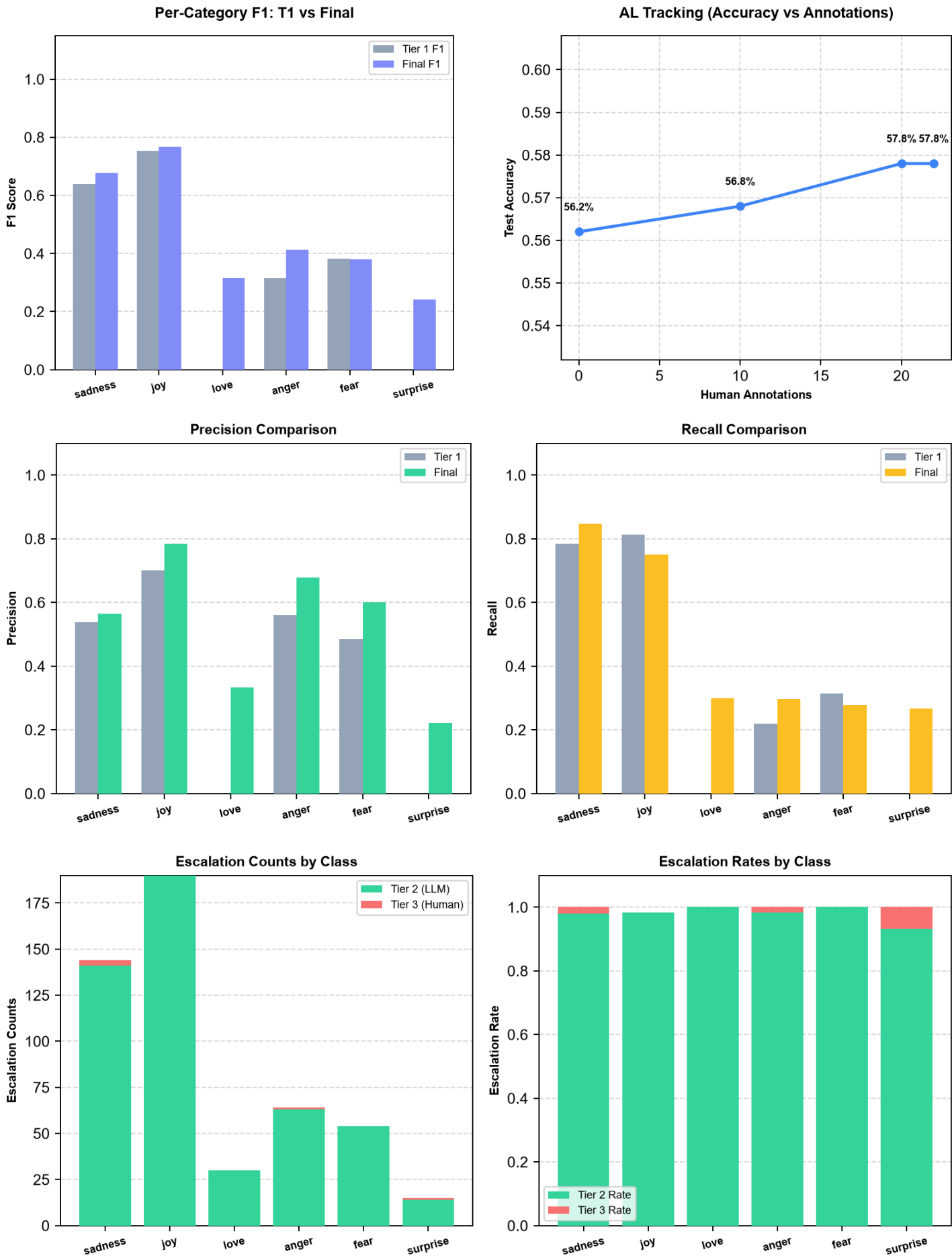


Efficiency Frontier Analysis

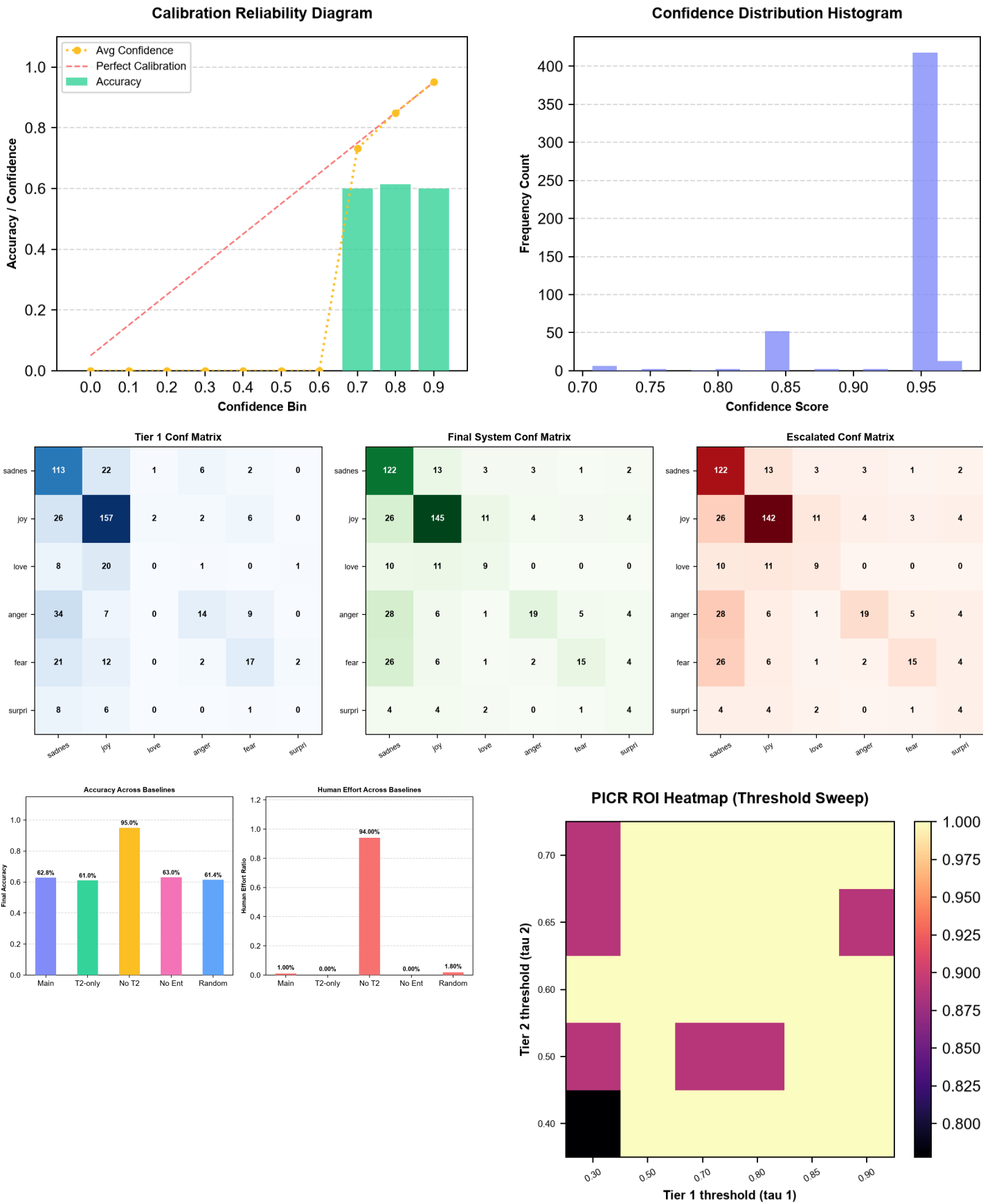




# Category & Escalation Profile: EMOTION



# Calibration & Validation Studies: EMOTION

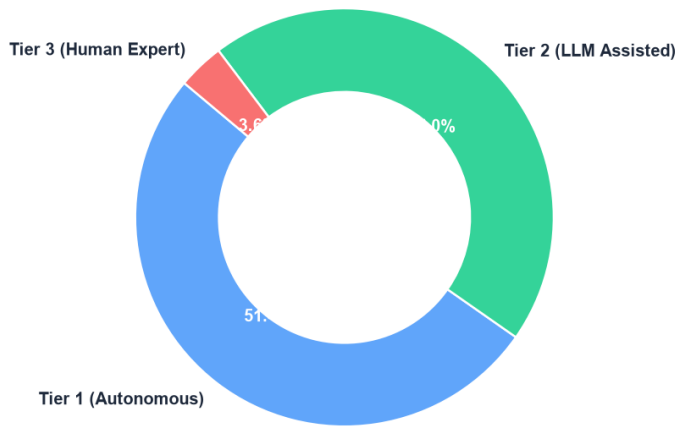


# Dataset Analysis: IMDB

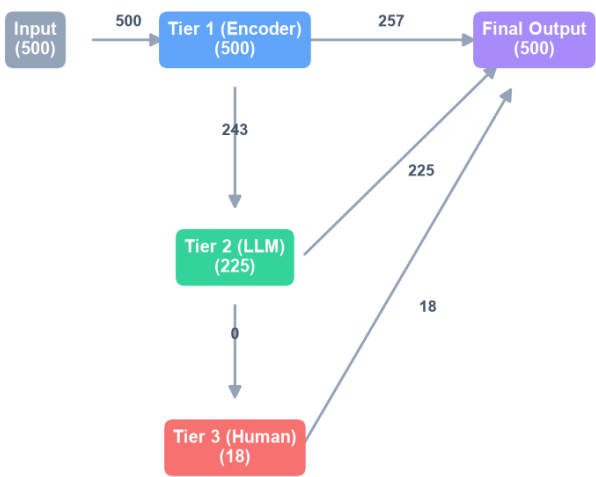
## Performance & Budget Summary Metrics

|                               |        |                              |              |
|-------------------------------|--------|------------------------------|--------------|
| Total Evaluated Samples:      | 500    | Human Effort Ratio:          | 3.60%        |
| Pre-AL T1 Accuracy:           | 79.40% | Post-AL T1 Accuracy:         | 79.40%       |
| Routed Final System Accuracy: | 87.40% | Final System F1 Macro:       | 0.8734       |
| Brier Score (Final):          | 0.2509 | Final System ECE:            | 0.0719       |
| Net Utility (Human-Effort):   | 0.0782 | Net Utility (Cost-Aware):    | 0.0767       |
| Average Compute Cost:         | 198.51 | Active Learning Escalations: | 25 sample(s) |

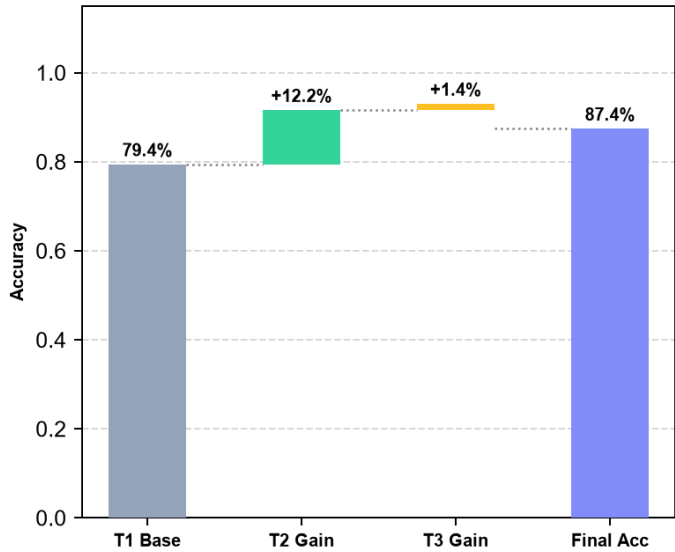
Request Routing Distribution



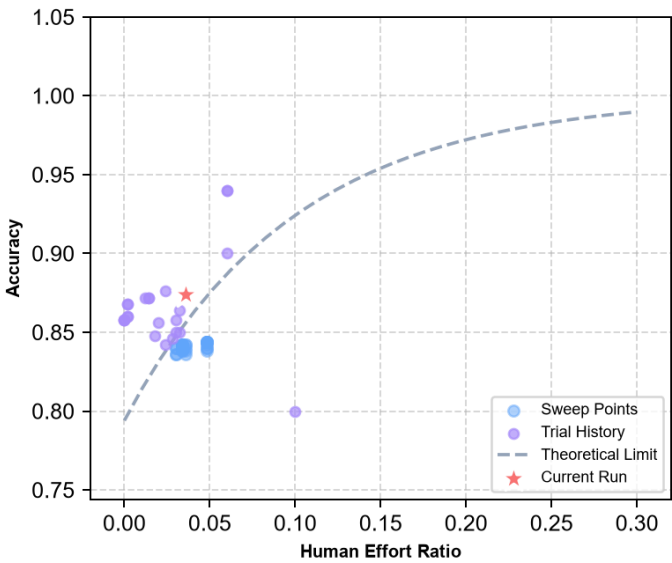
Request Flow Pipeline (Sankey)



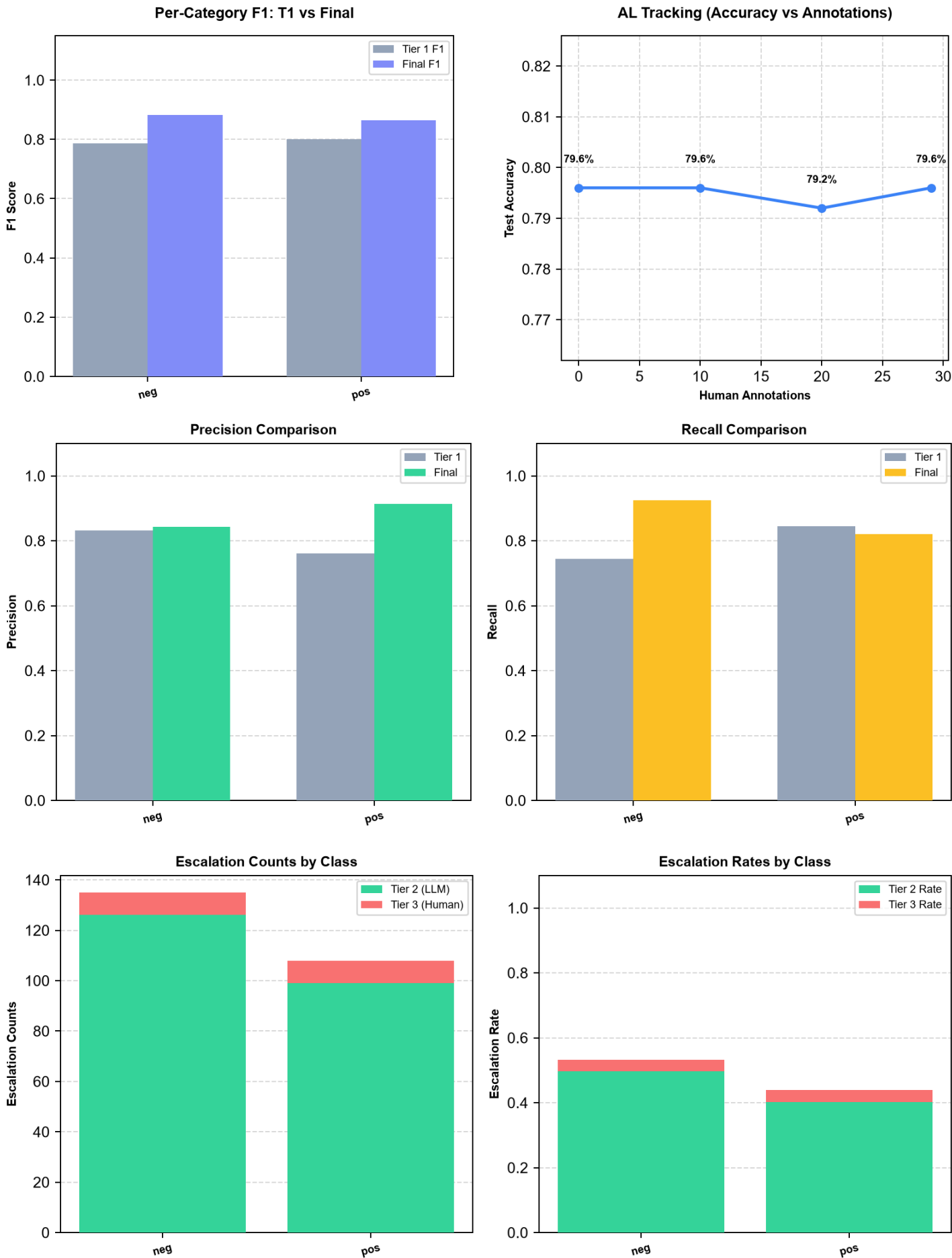
Accuracy Waterfall Study



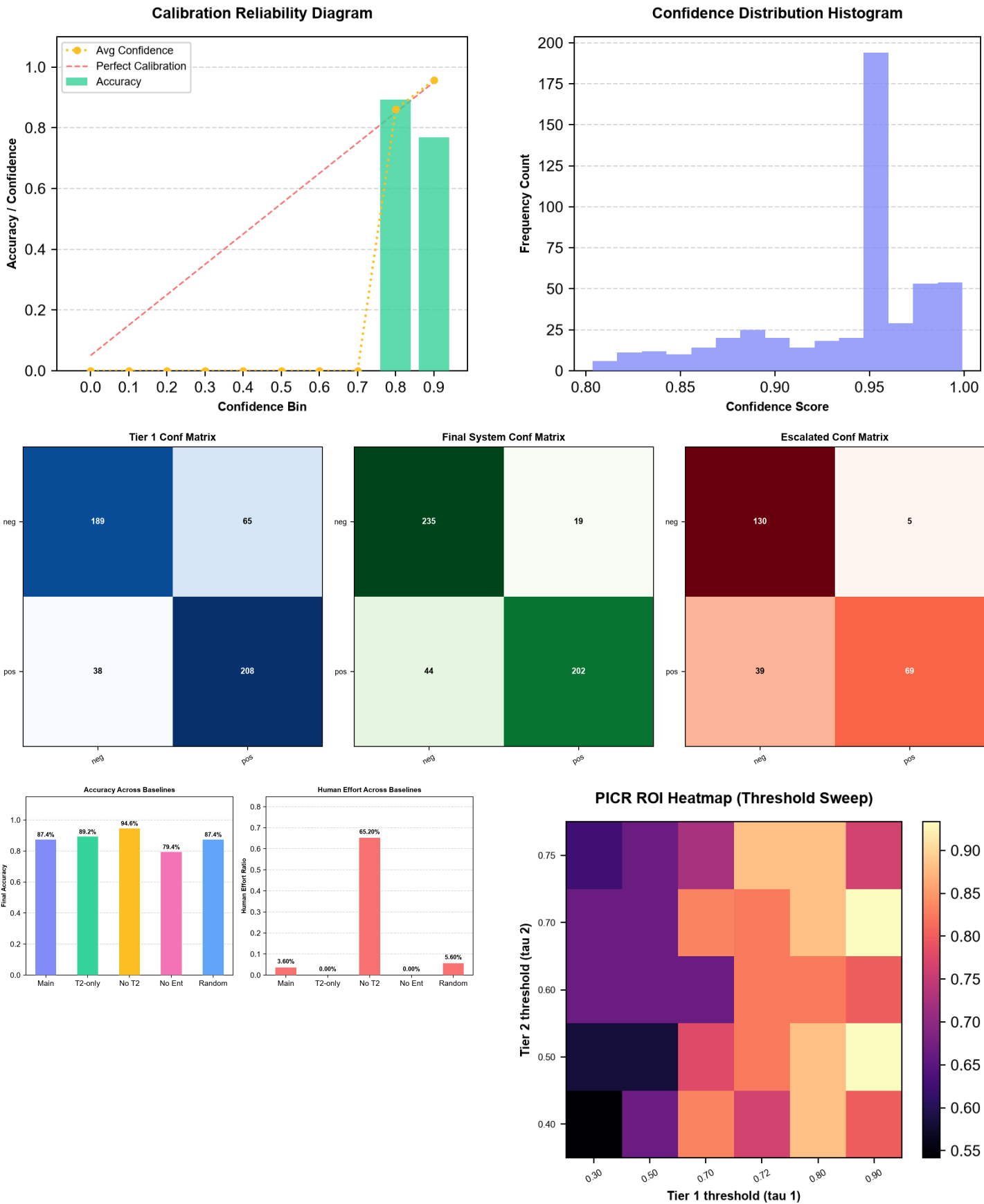
Efficiency Frontier Analysis



# Category & Escalation Profile: IMDB



# Calibration & Validation Studies: IMDB



# Cross-Dataset Comparison Synthesis

